

Microbial Biotechnology and Genetics

Science

May, 2026

Microbial Biotechnology and Genetics (A13838)

Short Title: Microbial Biotech. & Genetics
Faculty Science and Computing
Department Science
Cluster Biology
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Credits 5

Level: Introductory

Description of Module / Aims

The aim of this module is to introduce the principles of microbial molecular biology and regulation of gene expression. In vivo and in vitro techniques of bacterial genetics are also covered in this module. An introduction to basic fermentation is given and products of biotechnology are explored.

Programmes

MBIO-0006	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	2 / 4 / M
MBIO-0006	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	2 / 4 / M

Indicative Content

- Macromolecules and genetic information
- DNA structure & replication
- RNA & protein synthesis
- Regulation of enzyme activity & transcription (positive and negative control)
- Bacterial genetics: mutation & recombination
- In vivo & in vitro techniques of bacterial genetics
- Introduction to bioprocess design and control
- Bioreactors
- Genetic engineering and biotechnology

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Recall and explain biological information flow.
2. Describe the central dogma of genetics from DNA replication to protein synthesis.
3. Explain how cells control the expression of their genetic information.
4. Compare and contrast regulatory mechanisms in prokaryotes and eukaryotes.
5. Discuss mutation and genetic recombination.
6. Construct a flow diagram of a typical biotechnological process & describe bioreactor design suited to different applications.
7. Employ basic techniques of in-vitro DNA manipulations.

Learning and Teaching Methods

- A multiple modalities approach to teaching basic molecular concepts is used.
- The creation of multimodal texts/content provide opportunity for the students to use linguistic, visual and audio modes in order to experience, conceptualize, analyze and apply meaning.
- E-activities, online tutorials, digital storyboards, videos, e-forums, mindmaps and concept maps are some blended learning activities in the design of this module

- Face to face lectures provide opportunity for the students to conduct some of the learning activities such as concept maps & mindmaps
- Virtual lectures are created as 10-15 mins short digital storyboards. Students engage with this material prior to face to face sessions.
- Students participate in online MCQ quizzes after reviewing each virtual storyboard.
- Student reflection is conducted using group forum discussions.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
<i>Practical</i>	50%	6,7

Assessment Criteria

<40%: No understanding of microbial biotechnology and molecular genetics. Unable to carry out critical thinking or practical skills.

40%-49%: Able to understand basic concepts. May have some difficulty with critical thinking and problem solving. Adequate practical skills.

50%-59%: A good understanding of the material presented with demonstration of some critical thinking and problem solving skills. Good practical skills.

60%-69%: In addition, good critical thinking skills and showing a high level of competence and efficiency in theoretical and practical microbial biotechnology.

70%-100%: All of the above to an excellent level.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- "Microbial Biotechnology and Genetics module Moodle page." <http://moodle.wit.ie>

Supplementary Material

- Glazer, A. and H. Nikaido. *Microbial Biotechnology: Fundamentals of Applied Microbiology*. 2. Cambridge: Cambridge University Press, 2007.
- Klug, W. *Essentials of Genetics*. 7th Edition illustrated. USA: Benjamin Cummings, 2010.
- Madigan, M., J. Martinko, K. Bender, D. Buckley, D. Stahl and Brock T. *Brock Biology of Microorganisms*. 14th Edition. UK: Wiley, 2012.
- Waites, M., N. Morgan, J. Rocky and G. Higton. *Industrial Microbiology: An introduction*. UK: Wiley-Blackwell, 2002.

Requested Resources

- Room Type: Biology Lab

- EQUIPMENT: Blackboard
- Lecture Room: Loose Seated